

Final Project: Minimum-Energy Trajectory Optimization for Lunar Rover Navigation

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Problem Definition

A large proportion of future lunar missions would be enabled by autonomous rovers. However, the Moon's surface is covered with craters, steep slopes, and loose regolith that can trap or damage a rover — making naive straight-line traversal infeasible.

Our project team formulates this as a constrained trajectory optimization problem: given a lunar terrain, we need to find the path minimizing total energy consumption while avoiding craters and respecting slope limits imposed by rover stability requirements. The key tradeoff is that the flattest, safest path is rarely the shortest, and the shortest path may traverse terrain that is energetically expensive or physically dangerous.

We target a scenario relevant to micro rovers operating near the lunar south pole in support of Artemis surface missions, where limited battery capacity and sparse charging opportunities make energy-optimal navigation a critical operational requirement. [1][2]

Proposed Approach

We will model the lunar surface as a digital elevation map and derive terrain features relevant to rover mobility, such as local slope, roughness, and traversability masks. For terrain data, we plan to use Lunar Reconnaissance Orbiter products based on the Lunar Orbiter Laser Altimeter (LOLA) [4], which provides lunar topography, slope, and roughness products suitable for mobility analysis. We may use the open-source moon [9] package to extract crater cutouts from LOLA-based elevation data.

First, we will compute a feasible path using a terrain-aware global planner on the map, such as A* or Hybrid A*, with costs that penalize steep slopes and hazardous regions, following similar work done for lunar rover path-planning. [3]

Second, we will refine the path using continuous trajectory optimization to reduce estimated energy consumption while enforcing smoothness and rover motion constraints. The plan is to use a method such as direct collocation [6], because it is a standard and well-documented approach for trajectory optimization and is practical to implement using tools such as CasADi [7].

For the rover model, we will begin with a simple planar kinematic model with position and heading, together with bounds on turning rate and speed. Energy consumption will be approximated using a cost that captures path length, uphill travel, and control effort, following the general idea used in energy-aware planning on uneven terrain [8].

Measure of Success

We will evaluate success based on feasibility, energy savings, and efficiency. A solution must produce a valid path from start to goal while avoiding hazardous terrain and satisfying slope and rover motion constraints. We will compare our method against baseline planners using estimated energy, path length, maximum slope, smoothness, feasibility rate, and solve time. The project will be successful if our method consistently finds feasible trajectories and reduces estimated energy relative to the baselines without excessive computation time. [3][8]

Sketches of Solution

1. Extract a local lunar terrain cutout around a cratered region and convert it into an elevation grid. [4][9]
2. Compute derived layers such as slope and traversability. [4]
3. Choose start and goal positions on opposite sides of challenging terrain.
4. Run a global planner to obtain a collision-free and slope-feasible initial path. [3]
5. Parameterize that path as a sequence of waypoints or spline control points.
6. Solve a trajectory optimization problem that minimizes an energy objective, subject to rover kinematic constraints, bounds on control inputs, and terrain safety constraints. This follows the general structure of methods for optimal control. [6][7]
7. Compare the trajectory to the initial feasible path and other baselines. [3][8]

If time permits, we may extend the formulation to include more realistic terrain-dependent energy terms or illumination-aware constraints relevant to south-pole missions [3].

References

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